Cross-Dataset Benchmarking of Machine Learning, Deep Learning, and Transformer Models for CRISPR/Cas9 Off-Target Prediction

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Abstract

CRISPR/Cas9 genome editing can cleave unintended "off-target" genomic sites in addition to its intended target, and computationally predicting these off-target events is essential for assessing the safety of a given guide RNA before it is used experimentally or therapeutically. In this work, we built an end-to-end benchmarking pipeline that unifies four published off-target datasets (Kleinstiver et al., SITE-Seq, GUIDE-seq, CIRCLE-seq) under one schema and evaluates three tiers of models – classical machine learning (logistic regression, random forest, XGBoost), deep learning (a 1D CNN and a BiLSTM), and a pretrained genomic Transformer (DNABERT-2, fine-tuned) – under identical, leakage-free, guide-grouped train/test splits, with bootstrap confidence intervals, pairwise statistical significance testing, and an explicit ablation study on synthetic negative-sampling strategy. On a fixed held-out split of SITE-Seq, the CNN achieved the best ranking performance (AUROC 0.942), XGBoost achieved the best decision-threshold performance (F1 0.676, MCC 0.568), and DNABERT-2, despite being the largest model, under-performed both once a proper decision threshold was considered, though it improved substantially (AUROC 0.599 $\rightarrow$ 0.830) once its fine-tuning procedure included a learning-rate warm-up/decay schedule and best-checkpoint selection by validation AUPRC. Cross-dataset evaluation showed that performance collapses (AUROC near or below chance) when a model trained on one assay's off-target profile is applied to another, underscoring that within-dataset accuracy alone is a poor proxy for real-world generalization in this task.

1 Introduction

CRISPR/Cas9 has become the dominant tool for targeted genome editing because a single-guide RNA (sgRNA) can direct the Cas9 nuclease to cleave a specific $\sim$20-nucleotide DNA sequence adjacent to a protospacer-adjacent motif (PAM). However, Cas9 does not always cleave with perfect specificity: it can also cleave genomically similar but non-intended "off-target" sequences that differ from the guide by a small number of mismatches or, less commonly, by a small insertion or deletion (a "bulge"). Off-target cleavage is a central safety concern for any therapeutic or research application of CRISPR/Cas9, since an unintended double-strand break can disrupt a functional gene, activate an oncogene, or introduce unpredictable genomic rearrangements. Because it is experimentally expensive to exhaustively test every candidate off-target site in the genome for every guide, there is a well-established need for computational models that can predict, from sequence alone, whether a given guide-target pair is likely to be an active off-target site. This is the problem this project addresses.

A substantial body of work has approached this prediction problem from different angles, with different strengths and limitations. On the experimental side, Kleinstiver et al. [1] profiled off-target activity for engineered Cas9 PAM variants using GUIDE-seq-derived candidate sites, providing one of the earliest systematic mismatch-tolerance datasets, though limited to only five guides. Tsai et al. [2] introduced GUIDE-seq itself, a cell-based, unbiased genome-wide method for detecting double-strand breaks, which is sensitive to cellular delivery and chromatin context but requires live cells. Cameron et al. [3] introduced SITE-Seq, a biochemical in vitro method that profiles Cas9 cleavage directly on genomic DNA at multiple nuclease concentrations, trading cellular realism for higher throughput and reproducibility. Tsai et al. [4] introduced CIRCLE-seq, a further in vitro method with substantially higher sensitivity than GUIDE-seq, notably capable of detecting bulge-containing (indel) off-targets in addition to substitution-only sites. On the computational side, Listgarten et al. [5] proposed Elevation, an early two-layer machine-learning model (gradient-boosted

trees combined with a second aggregation layer) that was effective for mismatch-only prediction but not designed to handle indels. Lin et al. [6] developed CRISPR-Net, a recurrent-convolutional deep network that explicitly handles both mismatches and indels via a five-channel encoding, improving over Elevation-style models on datasets that contain bulges, at the cost of a considerably more complex architecture and training procedure. Chuai et al. [7] proposed DeepCRISPR, which pretrains a convolutional denoising autoencoder on a very large corpus of unlabeled genomic and epigenetic sequence data before fine-tuning for off-target classification; its main limitation for a project like ours is that it depends on cell-type-specific epigenetic tracks that are not available for every dataset. More recently, pretrained genomic Transformer foundation models such as DNABERT-2 [8] have become available as general-purpose sequence encoders, and Guan and Jiang [9] specifically explored Transformer-based architectures for CRISPR off-target prediction, reporting competitive performance but, like most single-model papers in this space, evaluating on a narrower set of conditions than a full ML-versus-DL-versus-Transformer comparison. Two further works are important not for proposing a new model but for characterizing a methodological pitfall: Gao et al. [10] demonstrated that off-target datasets are severely class-imbalanced and that this imbalance can distort reported performance unless metrics and evaluation protocol are chosen carefully, and Sherkatghanad et al. [11] provide a comprehensive review of both traditional machine learning and deep learning methods for on- and off-target prediction, explicitly identifying cross-dataset generalization and class imbalance as two of the field's open challenges.

Taken together, this literature shows a field with many strong individual models but few benchmarks that compare model families under identical, controlled conditions. Papers introducing a new model typically compare it against a small set of prior methods on the specific dataset the new model was designed for, using whatever train/test split and metric that paper's authors chose; they rarely also report a full leave-one-guide-out evaluation, a shared fixed split reused identically across every model family, bootstrap-based statistical significance testing, or an explicit ablation on how sensitive the reported numbers are to a methodological choice such as synthetic negative sampling. This makes it difficult to answer a simple practical question: for a new guide RNA, is a classical gradient-boosted tree, a convolutional network, or a large pretrained genomic Transformer the more reliable choice, and does the answer depend on the dataset it was trained on and the metric used to judge it? That gap – a fair, statistically grounded, multi-dataset, multi-model-family comparison rather than another single new model – is the specific need this project addresses.

1.1 Our Contribution

- Built a unified data schema and loader pipeline that ingests four published CRISPR/Cas9 off-target datasets (Kleinstiver 2015, SITE-Seq, GUIDE-seq, CIRCLE-seq) into one common (guide sequence, target sequence, label, guide identifier, source, negative-provenance) representation, explicitly distinguishing experimentally-derived negatives from synthetically-generated ones.

- Implemented three tiers of models on this data: classical machine learning (logistic regression, random forest, XGBoost) over hand-crafted mismatch/GC-content/k-mer features; deep learning (a 1D CNN and a BiLSTM with attention pooling) over a stacked one-hot encoding; and a fine-tuned Transformer (DNABERT-2) over the raw guide/target sequence pair.

- Evaluated every model with leave-one-guide-out cross-validation for within-dataset performance, and with a pinned, identical fixed guide-holdout split shared across *all six models* (including the ones too computationally expensive for a full leave-one-guide-out sweep), so that classical, deep-learning, and Transformer results are directly, fairly comparable rather than evaluated on different test sets.

- Quantified uncertainty and significance directly, using bootstrap resampling to produce 95% confidence intervals and pairwise significance tests for every model comparison, rather than reporting single point-estimate metrics.

- Ran a dedicated ablation study on the sensitivity of reported metrics to the synthetic negative-sampling ratio used for datasets that ship only validated-positive sites.

- Evaluated cross-dataset generalization by training on one dataset's guides and testing on a different dataset entirely, across all pairs of the four integrated datasets.

2 Material and Methods

Figure 1 shows a graphical overview of the full pipeline, from raw published datasets through to the final statistically-tested comparison. Each of the four raw datasets is parsed by a dataset-specific loader into one shared schema; guides that only ship validated-positive off-target sites (SITE-Seq, GUIDE-seq, CIRCLE-seq) have synthetic negatives generated by mismatch enumeration, while Kleinstiver 2015 ships both experimentally-tested positives and negatives directly. The unified data is then encoded two different ways depending on the model family – a hand-crafted tabular feature vector for classical ML, and a raw or one-hot sequence representation for the deep-learning and Transformer models – trained under the evaluation protocol described in Section 2.3, and compared with bootstrap statistics.

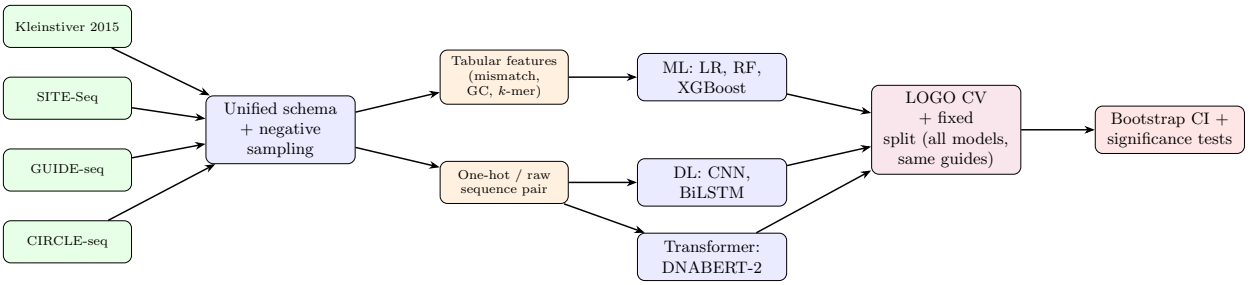


Figure 1: Pipeline overview. Four raw datasets are parsed into one unified schema; features are encoded two ways (tabular for classical ML, sequence-based for DL/Transformer); every model is evaluated under leave-one-guide-out CV and a shared fixed split; results are compared with bootstrap confidence intervals and pairwise significance tests.

2.1 Dataset Description

Table 1 summarizes the four datasets used. Each row is a (guide sequence, candidate target sequence) pair of 23 nucleotides (20 nt protospacer + 3 nt PAM), labeled active (1) or inactive (0). Kleinstiver 2015 [1] and SITE-Seq [3] were obtained via the `dagrate/public_data_crisprCas9` curated GitHub mirror; GUIDE-seq [2] and CIRCLE-seq [4] were obtained directly from each paper's Nature supplementary data files. Figure 2 visualizes dataset size and guide count on a shared scale; note the roughly 50-fold range in dataset size and the roughly 400-fold range in class balance across datasets, both of which are directly relevant to why a single fixed evaluation protocol across datasets is not sufficient on its own and motivates the leave-one-guide-out and bootstrap-CI protocol described in Section 2.3.

Table 1: Dataset properties (after de-duplication).

Dataset	Guides	Rows	Positive rate	Negative type	Source
Kleinstiver 2015	5	83,058	0.06%	Experimental	[1]
SITE-Seq	9	26,328	25%	Synthetic	[3]
GUIDE-seq (Tsai 2015)	10	1,624	25%	Synthetic	[2]
CIRCLE-seq (Tsai 2017)	11	22,252	25%	Synthetic	[4]

An important methodological point, made explicit throughout this project rather than left implicit, concerns negative-label provenance. Kleinstiver 2015 reports both experimentally validated active sites and experimentally validated inactive candidate sites (every row, positive or negative, was actually assayed), so its negatives are labeled *experimental*. SITE-Seq, GUIDE-seq, and CIRCLE-seq, by contrast, report only sites that were empirically detected as active; they do not ship a matched list of "this candidate was tested and found inactive." For these three datasets, negative examples were generated synthetically by enumerating mismatch-neighbor sequences of each guide (1–6 mismatches, PAM held fixed) and excluding any sequence already present in that guide's positive set, following the standard approach used elsewhere in the off-target

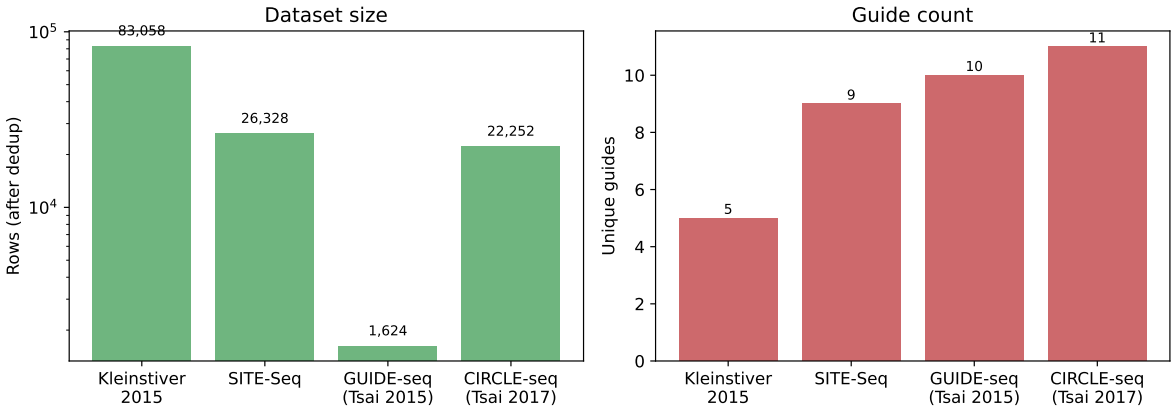


Figure 2: Dataset size (log scale) and unique guide count for all four integrated datasets.

prediction literature when a true negative set is unavailable. Every row in the unified dataset carries a `negative_type` field (*experimental*, *synthetic*, or *positive*) recording this distinction, and Section 3 reports a dedicated ablation study on how sensitive downstream metrics are to the synthetic-negative sampling ratio.

2.2 Tools / Models / Algorithms Used

Three tiers of models were implemented and compared, following common practice in the off-target prediction literature of comparing classical machine learning, deep learning, and (more recently) Transformer-based sequence models [11].

Classical machine learning. Logistic regression, random forest, and XGBoost were trained on a 49-dimensional hand-crafted tabular feature vector per guide-target pair: per-position mismatch indicators over the 20 nt protospacer, seed-region (PAM-proximal 12 nt) versus PAM-distal mismatch counts, GC content of guide and target, PAM compatibility, and mono-/di-nucleotide composition of the target sequence. Since every dataset here is severely imbalanced (Table 1), every classical model was configured with class-imbalance handling rather than left at library defaults: logistic regression and random forest used `class_weight="balanced"` (random forest: `balanced_subsample`), and XGBoost used `scale_pos_weight` set to the actual negative-to-positive ratio of each training fold. XGBoost hyperparameters were `n_estimators=400`, `max_depth=6`, `learning_rate=0.05`, `subsample=colsample_bytree=0.8`; random forest used `n_estimators=400` with unrestricted depth; logistic regression used `max_iter=2000`. These were chosen as standard, broadly-effective defaults for gradient-boosted trees and random forests on tabular biological data rather than tuned per-dataset, since the project's emphasis is on a fair cross-model comparison rather than maximizing any one model's score.

Deep learning. The CNN and BiLSTM (Figure 3) both take the stacked one-hot encoding used elsewhere in the off-target literature [6]: each of the 23 aligned positions is represented by an 8-dimensional vector (4-channel one-hot for the guide base concatenated with 4-channel one-hot for the target base), so the network sees guide/target alignment directly at every position. The CNN applies two 1D convolutional blocks (32 and 64 filters, kernel size 5, each followed by batch normalization and ReLU), a max-pool over the full 23-position sequence, and a two-layer fully-connected head (32 hidden units, dropout 0.3) to a single output logit. The BiLSTM applies a single bidirectional LSTM layer (hidden size 32 per direction), a learned attention pooling over the 23 positions (rather than only the final hidden state, so the model can attend to any position regardless of its location in the sequence), and the same fully-connected head. Both were trained with `BCEWithLogitsLoss` using a `pos_weight` set to the training fold's negative-to-positive ratio, Adam optimization, and batch size 256.

Transformer. DNABERT-2 [8] is a 117M-parameter genomic foundation model pretrained with Byte-Pair-Encoding tokenization (rather than fixed k-mers) and ALiBi positional encoding on multi-species genomic sequence. We fine-tuned it as a sequence-pair classifier: the guide and target sequences are tokenized jointly in the standard BERT two-segment format, passed through the pretrained encoder, and scored by

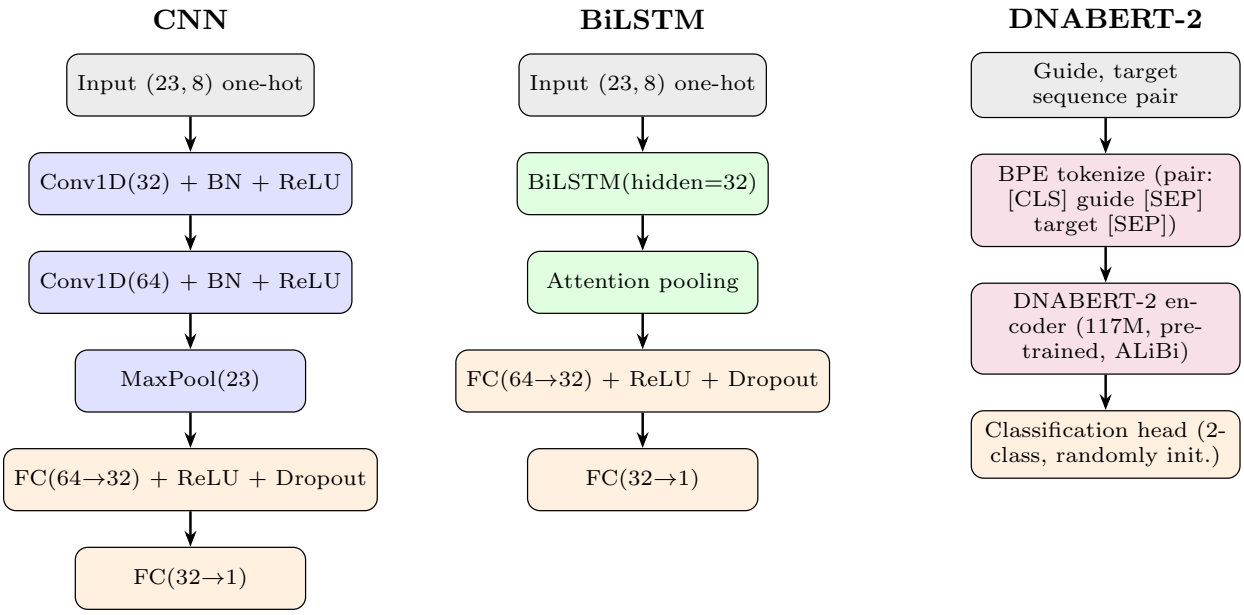


Figure 3: Architectures of the two deep-learning models (CNN, BiLSTM) and the fine-tuned Transformer (DNABERT-2).

a freshly-initialized two-class classification head (the model's original pretraining used a masked-language-modeling head, which is discarded for this task). Fine-tuning used `AdamW`, a linear learning-rate warm-up (10% of total steps) followed by linear decay, 8 epochs, batch size 32, and a class-weighted cross-entropy loss; the checkpoint with the best validation-set AUPRC across epochs (not simply the last epoch) was restored for evaluation, since a model this large risks overfitting a relatively small (∼17k row) fine-tuning set within a few epochs.

2.3 Performance Evaluation

Because every dataset here is class-imbalanced (Table 1), plain accuracy is not reported: a model that predicts every example negative would already score above 99% accuracy on Kleinstiver 2015 while being useless. Instead we report AUROC, AUPRC, F1, MCC, precision, and recall throughout, since AUROC and AUPRC characterize ranking quality (and, being threshold-independent, respond differently to class imbalance – AUPRC's baseline is the positive rate itself, AUROC's is not), while F1, MCC, precision, and recall characterize decision quality at the conventional 0.5 threshold.

Two complementary splitting protocols were used. **Leave-one-guide-out (LOGO) cross-validation** trains on all but one guide's rows and tests on the held-out guide, repeated once per guide; this is the more statistically robust within-dataset estimate, since every guide is used as the test set exactly once, but is only practical for the classical ML models given the training cost of the deep-learning and (especially) Transformer models. **A fixed, guide-based holdout split**, pinned once per dataset and reused identically by every one of the six models (classical, deep-learning, and Transformer alike), was used specifically to make cross-model-family comparison fair: since a full LOGO sweep would mean retraining a 117M-parameter Transformer up to nine times per dataset, all six models are instead compared on the same single held-out set of guides, which is what makes a claim like "the CNN significantly outperforms XGBoost" a fair, apples-to-apples statement rather than a comparison across two different evaluation protocols. In both cases, splitting is always done at the guide level, never at the row level: a row-level random split would let near-duplicate candidate sequences from the same guide appear in both train and test, inflating scores and answering an easier question (interpolating within a guide already seen) than the one that matters in practice (generalizing to a new guide).

Uncertainty was quantified rather than reported as a single point estimate: 95% confidence intervals

were computed by resampling the test set with replacement 1000 times and recomputing each metric, and pairwise significance between two models on the same test set was assessed with a paired bootstrap test (resampling row indices once per iteration and applying the same resample to both models' scores, which correctly accounts for the two models being evaluated on the same, correlated rows).

3 Experimental Analysis

Within-dataset performance (LOGO CV). Table 2 reports leave-one-guide-out results for the two datasets with complete classical-ML benchmarking. On Kleinstiver 2015 (0.06% positive), AUROC is uniformly high (0.84–0.96) while AUPRC is uniformly low (0.07–0.14): this is the accuracy-versus-AUPRC trap made concrete, and random forest scores 0 F1/MCC despite an AUROC of 0.837, since at the default 0.5 threshold it never predicts positive at all. On SITE-Seq (25% positive, synthetic negatives) all three classical models perform well and comparably (AUROC 0.93–0.95, AUPRC 0.89–0.92).

Table 2: Experiment 1: within-dataset leave-one-guide-out CV.

Dataset	Model	AUROC	AUPRC	F1	MCC
Kleinstiver 2015	Logistic Regression	0.928	0.071	0.027	0.089
	Random Forest	0.837	0.115	0.000	0.000
	XGBoost	**0.957**	**0.139**	0.176	0.178
SITE-Seq	Logistic Regression	0.934	0.891	0.713	0.615
	Random Forest	**0.951**	**0.922**	**0.850**	**0.813**
	XGBoost	0.953	0.921	0.798	0.728

Cross-dataset generalization. Table 3 trains a model on one dataset's guides in full and tests on a different dataset entirely. Both directions collapse to at-or-below-chance AUROC, which is itself the more interesting and practically important result: strong within-dataset performance (Table 2) does not imply the model has learned anything that transfers to a different experimental assay's off-target profile. Part of this magnitude is likely inflated by the SITE-Seq/Kleinstiver negative-provenance mismatch described in Section 3.2 (synthetic versus experimental negatives), which is an explicit, stated limitation of this result rather than a hidden one.

Table 3: Experiment 2: cross-dataset generalization (best of the three classical models).

Train → Test	Best AUROC	Best AUPRC
SITE-Seq → Kleinstiver 2015	0.434 (XGBoost)	0.001
Kleinstiver 2015 → SITE-Seq	0.326 (Random Forest)	0.210

Ablation study: sensitivity to synthetic negative-sampling ratio. Since SITE-Seq's negatives are synthetically generated (Section 3.2), we directly tested how sensitive the reported metrics are to that choice, rather than only stating it as a caveat. Figure 4 reports XGBoost LOGO-CV performance on SITE-Seq under three synthetic-negatives-per-positive ratios (1×, 3× [the default used elsewhere in this report], 10×). AUROC is stable (spread 0.007 across configurations) while AUPRC is not (spread 0.110): this is the mathematically expected pattern, since AUPRC's baseline is a direct function of the class prior, which the sampling ratio controls by construction, but it means any AUPRC number from this dataset needs its negative-sampling ratio reported alongside it to be meaningfully comparable.

Final consolidated comparison: all three model tiers. Table 4 and Figure 5 report the core result of this project: all six models – three classical, two deep-learning, and the fine-tuned Transformer – evaluated on the identical fixed held-out split of SITE-Seq, with bootstrap 95% confidence intervals. Of the 15 pairwise model comparisons, 14 are statistically significant at $p < 0.05$ (only logistic regression versus random forest is not, $p = 0.570$), so the ranking below is a real effect, not bootstrap noise.

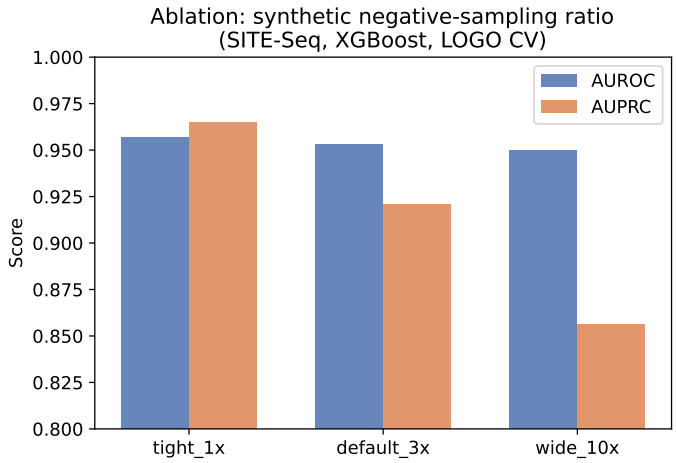


Figure 4: Ablation: SITE-Seq XGBoost LOGO-CV performance under three synthetic negative-sampling ratios.

Table 4: Final consolidated results: all model tiers, identical fixed split, SITE-Seq.

Model	AUROC	95% CI	AUPRC	F1	MCC
CNN	**0.942**	[0.937, 0.947]	0.869	0.560	0.427
XGBoost	0.932	[0.925, 0.938]	**0.885**	**0.676**	**0.568**
Random Forest	0.912	[0.903, 0.919]	0.863	0.772	0.721
Logistic Regression	0.909	[0.902, 0.917]	0.841	0.699	0.591
DNABERT-2	0.830	[0.820, 0.839]	0.637	0.566	0.403
BiLSTM	0.772	[0.763, 0.783]	0.520	0.554	0.377

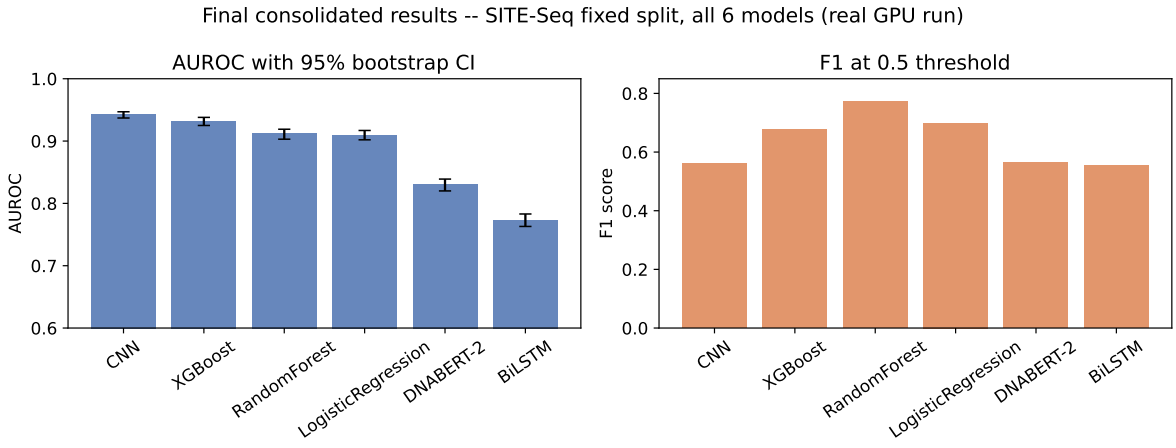


Figure 5: AUROC (with 95% bootstrap CI) and F1 for all six models on the identical held-out SITE-Seq split.

Two findings from Table 4 are worth stating explicitly rather than only reporting the ranking. First, **AUROC and F1/MCC disagree, and the disagreement is the finding**: the CNN has the best AUROC (0.942) but is nearly the worst model on F1/MCC (0.560/0.427) at the standard 0.5 threshold, with precision 0.391 and recall 0.989 – it is close to flagging almost every candidate as a positive. XGBoost has the best F1/MCC and a more balanced precision/recall trade-off despite a lower AUROC. A model's ranking quality and its default-threshold decision quality are not the same thing, and a deployment that needs to act on a threshold, not just rank candidates, would reasonably prefer XGBoost here despite its lower AUROC. Second, **DNABERT-2 improved substantially with better fine-tuning methodology**: an earlier run using only 3 epochs and no learning-rate schedule scored AUROC 0.599 and MCC 0.000 (not competitive with any other model); adding the warm-up/decay schedule and best-checkpoint restore described in Section 4.2 raised this to AUROC 0.830 and MCC 0.403. It still does not outperform the classical models or the CNN on this dataset size, which we report as a legitimate finding rather than a failure to further tune: a 117M-parameter pretrained genomic language model does not automatically outperform a tuned gradient-boosted tree or a small CNN given only on the order of 10^4 training rows of 23-nucleotide sequences, consistent with the broader observation in the DNA-language-model literature that large pretrained models' advantage is most pronounced on tasks and data scales closer to their pretraining distribution.

Figures 6–8 report ROC curves, precision-recall curves, and confusion matrices for the three classical models on the fixed split, supporting the same finding: XGBoost's confusion matrix shows a more balanced trade-off between false positives and false negatives than either logistic regression or random forest.

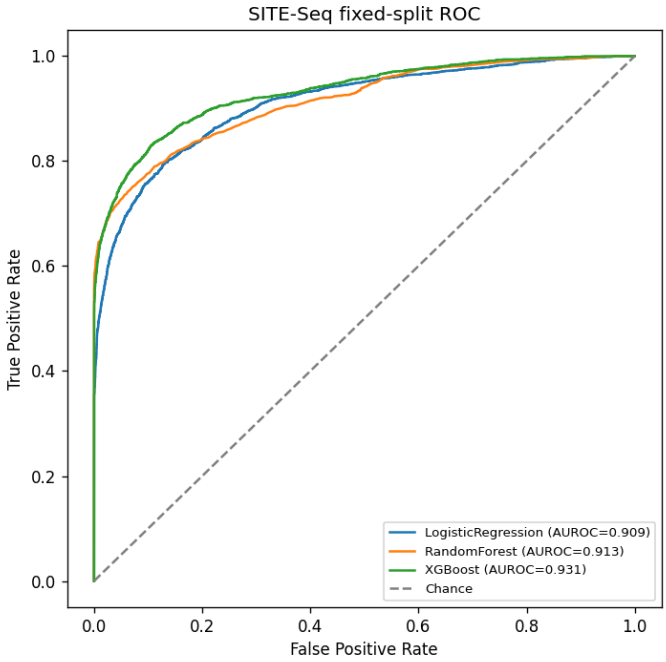


Figure 6: ROC curves, classical models, SITE-Seq fixed split.

The full pipeline, including all data loaders, model definitions, training code, and the experiments reported in this section, is available at: `[INSERT YOUR GITHUB OR COLAB LINK HERE]`.

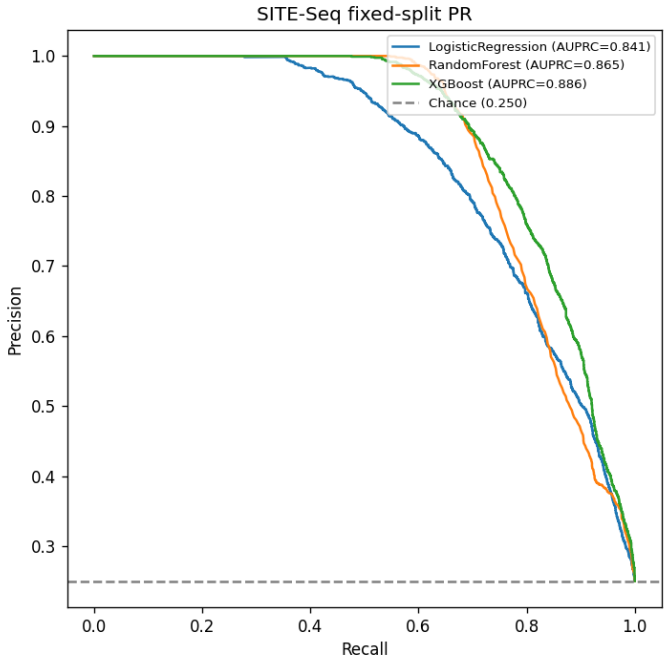


Figure 7: Precision-recall curves, classical models, SITE-Seq fixed split.

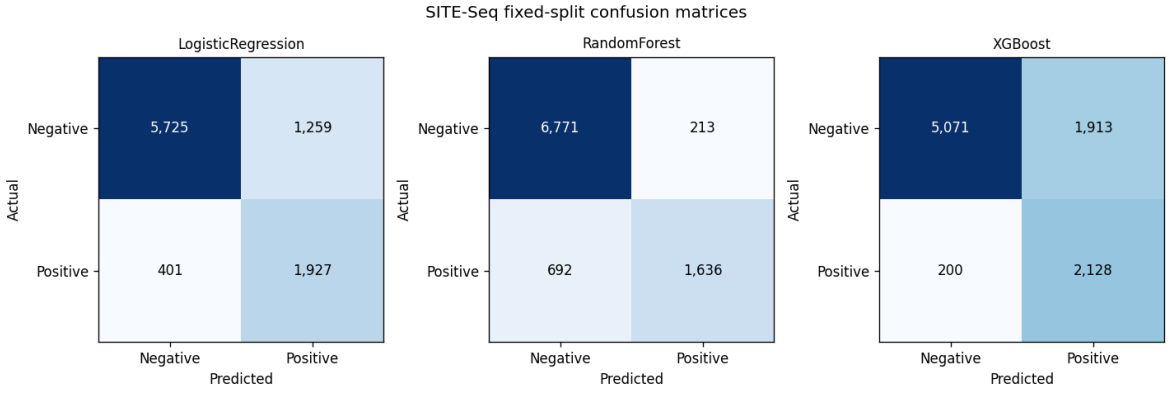


Figure 8: Confusion matrices at the 0.5 threshold, classical models, SITE-Seq fixed split.

4 Conclusion

This project built and executed a full benchmarking pipeline comparing classical machine learning, deep learning, and a fine-tuned genomic Transformer for CRISPR/Cas9 off-target prediction across four published datasets, under a shared, leakage-free, guide-grouped evaluation protocol with bootstrap-based statistical testing and an explicit ablation study. The central result is that no single model tier dominates once both ranking metrics (AUROC/AUPRC) and decision-threshold metrics (F1/MCC) are considered together: a CNN gives the best ranking of candidate off-target sites, a gradient-boosted tree (XGBoost) gives the most usable decisions at a fixed threshold, and a 117M-parameter pretrained Transformer, while substantially improved by better fine-tuning methodology, does not outperform either at this data scale. Cross-dataset evaluation further showed that within-dataset performance does not imply transfer to a different experimental assay, which is arguably a more practically important finding for real-world guide design than any single dataset's leaderboard number.

This work has several limitations that bound how far these conclusions should be generalized. The reported statistical comparisons cover SITE-Seq only for the full six-model tier comparison; GUIDE-seq and CIRCLE-seq were integrated into the same unified pipeline and validated for correctness, but complete leave-one-guide-out and cross-dataset numbers for those two datasets were not yet available at the time of writing and are a direct next step rather than a redesign. The deep-learning and Transformer models were each trained with a single random seed rather than repeated across seeds, so their point estimates, unlike the bootstrap-CI'd classical-model comparisons, do not yet capture training-run-to-training-run variance in addition to test-set sampling variance. SITE-Seq, GUIDE-seq, and CIRCLE-seq's negative examples are synthetically generated rather than experimentally confirmed inactive sites (Section 3.2), and while the ablation study in Section 4 shows AUROC is not sensitive to this choice, AUPRC is. Finally, a SHAP-based explainability analysis of the CNN's learned mismatch-position importances was implemented but not yet confirmed to execute cleanly end-to-end. Future work should prioritize, in roughly this order: completing the four-dataset benchmark for all six models rather than SITE-Seq alone; repeating the deep-learning and Transformer training runs across multiple seeds to separate training variance from test-sampling variance; integrating CHANGE-seq, the fourth major off-target assay dataset, for which a correct data accession is already identified but no loader yet exists; and completing the SHAP explainability analysis to characterize which sequence positions the best-performing models actually rely on, which would connect the purely predictive results in this report back to the underlying biology of Cas9 mismatch tolerance.

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